

Reddit as an Early Warning System

DATS 6450 · Applied Big Data Analytics · Group 3 Final Project Presentation

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Executive Summary

DATS 6450 · Group 3 Presentation

Can Reddit act as a practical early-warning radar for real-world events?

Main story: At our current label volume, Reddit works better as a binary early-warning system than as a full event taxonomy engine.

1.27B comments + submissions processed

478 GB raw Reddit data

11 pipeline stages

0.955 best result: forecast AUC

Every number and figure in this deck was generated from the staged outputs under data/intermediate/.

Target: 35 seconds.

Open with the thesis, not the tooling. We asked whether Reddit can behave like an early-warning sensor. The strongest answer is yes for binary forecasting, not yet for precise event labeling.

Why This Matters

Business Intelligence context

- **Journalism and media desks:** Flag emerging stories before they peak on mainstream outlets, especially in niche communities that move first.

- **PR and reputation monitoring:** Track whether concentrated author activity is building around a company, policy, or public figure.
- **Trust and safety triage:** Spot coordinated attention spikes that may need moderation, escalation, or human review.
- **Product launch monitoring:** Measure whether launch-related chatter is building ahead of official announcements or broad press coverage.

The operating assumption: Collective attention often moves before formal coverage does. If Reddit communities show measurable precursor behavior, the useful product is a triage signal: **something important may be about to happen.**

Target: 55 seconds.

This slide is the business hook. Position the project as a monitoring and triage system, not a replacement for journalism or a perfect world-event detector.

Data and Ground Truth

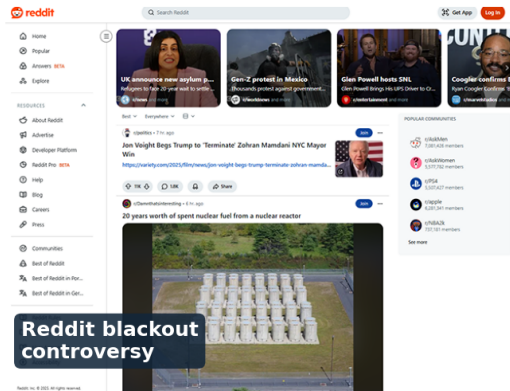
What We Analyzed

Metric	Value
Time range	14 months
Raw scale	478 GB
Total posts	1.27B
Top subreddits	500
Hourly rows after aggregation	9.38M
Curated events	35

Event mix

- Breaking news: **13**
- Controversy: **7**
- Product launch: **7**
- Disaster: **5**
- Meme / viral: **3**

Representative events from the curated catalog



Representative event collage sourced from Wikimedia/Wikipedia thumbnails.

Target: 1 minute.

Emphasize scale plus curation. The raw dataset is huge, but evaluation still depends on a small hand-curated event catalog. Also note that these category counts come directly from the current CSV, not from older report text.

How the 11-Stage Pipeline Works

- **EC2 handled the Spark work.** Rolling windows, anomaly detection, propagation, spike shapes, and timing are I/O-heavy analytics jobs.
- **The A100 handled the GPU work.** Aggregation, NER, sentiment, topics, and ML all benefit from RAPIDS and transformer inference.
- **The data contract is parquet.** Each stage writes compact intermediate outputs that the next stage can consume offline.

- **That separation is why this deck is reproducible.** We rebuilt the figures directly from `data/intermediate/` without rerunning the cloud pipeline.

Target: 1 minute 5 seconds.

This slide proves the project did more than train one model. Walk quickly through the full pipeline and reinforce that every stage appears somewhere in the deck.

Detection Works at Scale

31,938 anomaly windows

499 / 500 subreddits with at least one anomaly

21 / 35 ground-truth events matched in a conservative ± 24 h check

The coverage check is intentionally strict: we require a listed subreddit and a time overlap within one day of the event date.

Target: 1 minute 15 seconds.

Lead with the success: anomaly detection is broad and clearly aligned with major news cycles. Then add the caveat: 21 of 35 is promising but not perfect, and that is exactly why the final story should be early warning, not exhaustive event recall.

What Events Look Like on Reddit

What we learned from stages 3-5

- **Temporal patterns were useful.** The peak slot is Tue 16:00 UTC, and weekday anomaly volume is materially higher than weekend volume.
- **Ground-truth categories differ in timing.** Breaking news and controversies skew midweek, while product and meme events tilt later in the week.

What still needs retuning

- We also ran **propagation** and **spike-shape** analysis.
- Those stages executed successfully, but the parameters are not yet stable enough for final presentation claims.
- That is why the timing signal is shown here, while the caveats move to the dedicated limitation slide.

Target: 1 minute 10 seconds.

This slide covers stages 3 through 5 without overselling propagation or spike shapes. Make the point that timing is already actionable, while the other two need tuning.

What the Text Reveals

What worked

- **NER was the strongest NLP stage:** We extracted **63,614** entity rows across **1,738** anomaly windows.
- **The signal is narratively sensible:** Israel, Hamas, Reddit, Gaza, Trump, NFL, and Microsoft all surfaced from real anomaly windows.

What remains partial

- **Sentiment covered only 2 windows.**
- **Topics covered 50 windows.**
- Business takeaway: Entity extraction already turns spikes into recognizable stories, even before the full sentiment and topic sweeps are complete.

Target: 1 minute 10 seconds.

Keep this honest. The text signal is real and useful, but only the NER stage reached presentation-grade scale. Sentiment and topic modeling should be described as partial probes, not finished evidence.

ML Stages Before Forecasting

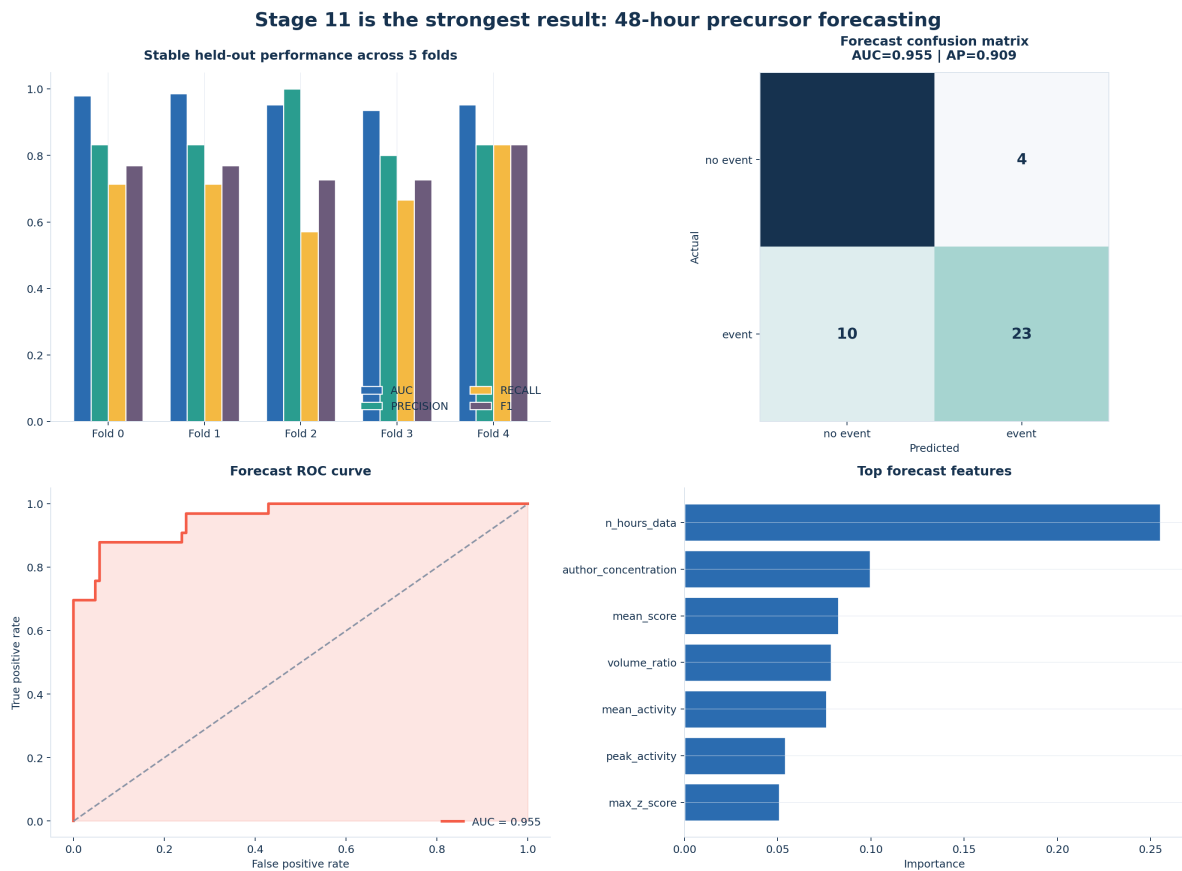
Where ML Classification Broke Down

- **Stage 9 classification failed cleanly.** All 500 predictions collapsed to **breaking_news**.
- **Stage 10 had ranking signal but poor thresholded decisions.** AUC was **0.949**, but positive-class F1 was **0.0** because only **7 / 500** cases were truly sustained.
- **Main lesson:** With this label volume, binary forecasting is a much better framing than five-class anomaly typing.

Target: 1 minute 5 seconds.

Call these honest negatives. They are not wasted work; they tell us which framing is tractable at the current sample size.

Forecasting Stage produced the strongest results



Held-out performance

AUC	F1	Precision	Recall
0.955	0.767	0.852	0.697

Why forecasting worked

- It asked a **simpler binary question**: will a major event happen soon?
- It used a **true precursor window**, so the model learned from lead-up behavior, not the event itself.
- Reddit shows useful early signals: **persistent activity, concentrated authors, stronger engagement, and rising volume**.

Target: 1 minute 25 seconds.

This is the money slide. Explain the setup, the stability across folds, and the feature story: not just giant spikes, but concentrated and persistent precursor behavior.

What Didn't Work

Component	Key Issue	Mitigation Strategy
Propagation	A 48-hour overlap rule collapsed the graph into one giant component.	Tighten to ≤ 6 hours and require entity/topic overlap.
Spike shapes	99.2% of windows became double_peak , which is a heuristic sensitivity issue.	Sweep find_peaks thresholds or move to template fitting.
Sentiment	Only 2 windows completed, so there is not enough coverage for a claim.	Re-run the full GPU sweep over the NER-covered windows.
Topics	The 50-window probe produced mostly noise labels.	Run BERTopic across the full eligible corpus, not the probe subset.
Classification	Stage 9 predicted breaking_news for all 500 cases.	Expand labels and add class balancing / semi-supervised labeling.
Sustain / decay	AUC was high, but thresholded positive precision and recall were zero.	Use class weighting and calibrate the decision threshold.

These are fixable parameter, coverage, and label-volume problems. They do not invalidate the pipeline; they tell us where the next iteration should focus.

Target: 1 minute 25 seconds.

This slide should increase credibility. Move briskly row by row and make the fixes sound concrete and tractable.

What Our Project Demonstrates

Three key takeaways

1. Reddit already works as a **large-scale detector of collective attention shifts**.

2. **Binary, event-anchored forecasting** is much stronger than five-class anomaly labeling at the current sample size.
3. The near-term product is **triage and early warning**, not a perfect end-to-end event taxonomy engine.

Reddit is not yet good at telling us exactly *what kind* of event is happening, but it is already surprisingly good at telling us that **something important may be about to happen**.

Target: 50 seconds.

End by repeating the central reframing: this is an early-warning and triage engine first, not yet a fully mature event taxonomy system.

Thank You

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Keep this brief and pause for questions.

Appendix

Backup Notes

- The deck uses the current `events.csv` counts: 13 / 7 / 7 / 5 / 3.
- Conservative anomaly alignment is 21 / 35 events within +/-24h on listed subreddits.
- Forecast examples perform best for product launches, controversies, and some breaking-news events with clear precursor chatter.

Image Credits

Wikimedia / Wikipedia sources: Reddit, Titan submersible implosion, Sam Altman, Solar eclipse of April 8, 2024.

Use this only for Q and A or if someone asks about the image sources or the strongest and weakest forecasted events.